

FENGNING, China

WMO: 543080

Lat: 41.2018N

Lon: 116.6343E

Elev: 2415

StdP: 13.46

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -3.7	(c) -0.5	(d) -23.3	(e) 1.7	(f) 7.4	(g) -20.1	(h) 2.0	(i) 8.8	(j) 27.7	(k) 11.3	(l) 24.9	(m) 11.6	(n) 3.9	(o) 320	(p) 0.502

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.0	89.6	66.0	86.8	65.9	84.3	65.9	73.9	82.1	72.2	80.0	70.5	78.3	8.2	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.4	126.9	78.8	69.8	119.9	77.3	68.0	112.6	75.6	39.4	82.2	37.8	80.2	36.2	78.4	81.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.4	19.9	16.8	-9.1	95.1	3.4	3.8	-11.6	97.8	-13.5	100.0	-15.4	102.1	-17.8	104.9
			-9.9	76.2	3.1	2.0	-12.2	77.7	-14.0	78.9	-15.8	80.0	-18.1	81.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	45.9	14.5	21.5	34.7	49.4	60.8	68.9	73.6	70.8	60.8	47.0	30.5	17.2	
	DBStd	21.71	6.09	7.34	8.09	7.80	6.15	4.90	4.00	4.44	6.08	6.80	7.82	6.23	
	HDD50	4233	1102	799	479	103	4	0	0	0	3	139	586	1018	
	HDD65	7606	1567	1219	941	469	155	20	1	7	150	558	1036	1483	
	CDD50	2746	0	0	3	86	340	568	730	645	328	46	0	0	
	CDD65	644	0	0	0	2	26	138	266	187	25	0	0	0	
	CDH74	6033	0	0	1	49	563	1433	2200	1483	300	4	0	0	
	CDH80	1826	0	0	0	5	146	471	751	400	53	0	0	0	
Wind	WSAvg	5.9	6.1	6.4	7.3	7.6	7.4	5.6	5.0	4.5	4.8	5.5	5.4	5.8	
Precipitation	PrecAvg	18.2	0.0	0.1	0.2	0.6	1.3	2.9	5.2	4.8	2.0	0.7	0.2	0.0	
	PrecMax	25.9	0.6	0.9	0.9	2.4	3.2	6.4	10.6	10.1	4.5	1.8	1.1	0.3	
	PrecMin	11.4	0.0	0.0	0.0	0.0	0.1	1.0	1.6	2.3	0.5	0.0	0.0	0.0	
	PrecStd	3.6	0.1	0.2	0.2	0.5	0.8	1.3	2.2	2.3	1.1	0.5	0.3	0.1	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	38.1	48.9	67.5	79.4	89.0	93.0	95.1	90.6	85.2	73.4	57.3	40.4	
		MCWB	27.7	34.7	45.4	53.8	59.8	63.1	68.3	69.1	63.0	54.7	42.7	29.7	
	2%	DB	33.3	44.0	60.2	74.3	84.1	88.6	90.1	86.9	80.8	68.1	52.1	35.8	
		MCWB	24.7	32.0	42.2	51.6	58.3	63.0	68.5	67.7	62.0	52.5	39.6	27.6	
	5%	DB	30.0	40.3	55.2	70.1	80.2	85.3	87.0	84.3	77.3	64.2	48.1	32.4	
		MCWB	22.5	29.9	39.5	50.3	56.4	62.8	68.2	67.6	60.2	50.1	37.6	25.1	
	10%	DB	26.8	36.3	50.5	66.0	76.5	82.1	84.2	81.8	73.9	60.6	44.2	29.2	
		MCWB	20.3	27.6	36.6	47.7	54.7	62.4	68.7	67.2	59.2	48.3	35.4	22.9	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	28.4	36.9	47.4	59.2	65.6	71.2	76.5	75.7	69.2	59.3	44.4	31.2
			MCDB	37.1	45.4	62.7	71.1	80.6	82.4	85.2	84.3	78.9	67.5	51.5	39.4
2%		WB	25.1	32.7	43.4	55.3	62.1	68.7	74.6	73.4	66.2	55.8	41.2	27.7	
		MCDB	32.8	42.2	58.4	70.1	76.4	79.0	83.0	81.4	75.5	63.7	49.8	35.1	
5%		WB	22.7	30.3	40.2	52.0	59.9	67.0	73.1	71.8	63.9	53.1	38.6	25.5	
		MCDB	29.6	40.0	53.5	65.9	72.9	77.5	81.0	78.9	72.5	60.5	47.4	31.8	
10%		WB	20.6	27.7	37.2	49.3	57.6	65.4	71.5	70.0	61.6	50.3	35.6	23.2	
		MCDB	26.6	36.2	49.0	62.9	70.5	76.1	79.5	77.2	69.8	57.9	43.7	29.0	
Mean Daily Temperature Range		5% DB	MDBR	21.8	23.3	23.7	25.0	25.6	22.5	19.0	20.2	22.9	22.5	21.2	21.1
			MCDBR	26.4	29.8	31.9	32.5	34.2	30.3	25.9	25.1	28.3	27.9	26.4	25.5
	MCWBR		19.5	20.2	18.5	15.7	13.9	10.4	8.8	9.5	11.7	14.3	15.8	18.4	
	5% WB	MCDBR	25.7	28.3	30.3	26.4	25.0	20.7	17.1	17.2	20.8	22.5	24.7	24.6	
		MCWBR	19.2	19.2	18.0	13.5	11.6	9.2	7.9	8.3	10.5	13.1	15.4	18.0	
Clear-Sky Solar Irradiance	taub		0.309	0.381	0.417	0.486	0.541	0.644	0.616	0.578	0.517	0.449	0.347	0.316	
	taud		2.283	2.070	2.014	1.896	1.793	1.643	1.749	1.810	1.882	1.979	2.223	2.274	
	Ebn at Noon		264	254	260	250	239	215	220	223	226	226	244	249	
	Edh at Noon		29	42	49	60	68	79	71	65	56	45	30	27	
All-Sky Solar Radiation	RadAvg		812	1088	1483	1743	1943	1757	1656	1618	1411	1110	818	698	
	RadStd		56	92	122	117	122	159	112	102	84	79	77	45	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (6 neighbors)	N/A	-1.66	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page